Stochastic Processes and Machine Learning in Commodity Price Forecasting: Insights from the Mitsui & Co. Commodity Prediction Challenge

Dalton Omondi

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Abstract

Commodity markets exhibit profound volatility, shaped by stochastic dynamics and multifaceted global influences. This paper advances the modeling of commodity returns through a dual lens: theoretical stochastic processes and practical machine learning pipelines. Inspired by the 2025 Mitsui & Co. Commodity Prediction Challenge on Kaggle, we formalize processes like Geometric Brownian Motion (GBM), Ornstein-Uhlenbeck (OU), and jump-diffusion models via stochastic differential equations (SDEs). We then integrate empirical insights from a comprehensive ML baseline—employing ensemble tree boosters (LightGBM, XGBoost, CatBoost) with cross-validation and feature engineering—trained on the competition's multi-asset dataset from London Metal Exchange (LME), Japan Exchange Group (JPX), U.S. stocks/ETFs, and Forex markets. Exploratory data analysis (EDA) reveals volatility clustering and mean reversion, aligning stochastic assumptions with data patterns. The ML pipeline achieves RMSE 0.029-0.030 across 424 targets, emphasizing stability via the challenge's Sharpe-variant metric. This research contributes a hybrid framework blending SDE simulations with tree ensembles for robust, low-variance forecasts, with code validated on competition data.

Keywords: Stochastic Processes, Geometric Brownian Motion, Ornstein-Uhlenbeck, Jump-Diffusion, Ensemble Learning, LightGBM, XGBoost, CatBoost, Time-Series Forecasting, Kaggle Competition

1 Introduction

1.1 The Imperative of Commodity Price Prediction

Global commodity prices drive economic cycles, yet their unpredictability—fueled by supply shocks, currency fluctuations, and geopolitical tensions—poses significant risks to traders and policymakers. The Mitsui & Co. Commodity Prediction Challenge (launched July 24, 2025; entry deadline September 29, 2025) provides a rigorous framework for addressing these forecasting challenges. Participants were tasked with predicting 424 price-difference series from paired assets, using historical data from the London Metal Exchange (LME), Japan Exchange Group (JPX), U.S. stocks/ETFs, and Forex markets [1]. Targets represent returns differences, evaluated via a Sharpe-ratio variant: mean Spearman correlation divided by standard deviation, prioritizing stable long-term predictions over short-term accuracy.

The dynamic behavior of commodity prices has been extensively modeled using stochastic processes, with Geometric Brownian Motion emerging as a particularly effective model for crude oil prices according to recent research [2]. Simultaneously, machine learning (ML) excels at pattern extraction from high-dimensional data. This paper synthesizes both approaches: deriving stochastic differential equations (SDEs) for theoretical grounding and dissecting an ensemble ML pipeline for practical deployment, with inspiration drawn from the winning solution that achieved a Sharpe ratio of 1.341129 in the Mitsui challenge [1].

1.2 Objectives and Structure

Primary Objectives:

- Derive and apply stochastic models to commodity price dynamics, emphasizing empirical validation.
- Analyze an end-to-end ML pipeline for the competition, including training outputs and inference procedures.
- Propose hybrid methodologies integrating stochastic simulations with machine learning for enhanced forecast stability.

The paper is structured as follows: Section 2 covers the mathematical foundations of stochastic processes. Section 3 applies these models to financial markets. Section 4 details the Kaggle competition framework. Section 5 presents exploratory data analysis. Section 6 examines the machine learning pipeline. Section 7 proposes integrated approaches. Section 8 concludes with findings and future research directions.

2 Literature Review

The forecasting of commodity prices has evolved through various methodological approaches. Traditional stochastic process models, particularly Geometric Brownian Motion (GBM), have demonstrated effectiveness in capturing the dynamic behavior of crude oil prices, with empirical studies showing strong alignment between simulated prices and market observations [2]. The unit root tests applied to crude oil price time series further confirm their stationarity, reinforcing the applicability of these continuous-time stochastic processes.

Recent advances incorporate decomposition techniques to enhance forecasting accuracy. Makaryev and Garashchuk (2020) demonstrated that decomposing commodity time series into several stochastic flows using singular spectral analysis significantly improves the forecast quality of ARIMA models for Brent crude oil, gold, and corn prices [4]. However, their findings also highlighted challenges in maintaining parameter stability across different sample periods, particularly when implementing trading strategies in volatile commodity markets.

Machine learning competitions have driven innovation in feature engineering and ensemble methods. The winning solution for the Mitsui Commodity Prediction Challenge achieved a 941.8% improvement over the baseline approach through comprehensive feature engineering, GPU-accelerated XGBoost with hyperparameter optimization, and intelligent ensemble weighting [1]. Similar approaches have proven effective in other financial forecasting domains, with feature engineering emerging as a critical differentiator in Kaggle competitions [5].

The integration of stochastic modeling with machine learning represents a promising frontier, potentially combining the theoretical rigor of SDEs with the pattern recognition capabilities of ensemble tree methods. This paper contributes to this emerging hybrid approach through empirical validation on a comprehensive multi-asset dataset.

3 Mathematical Foundations of Stochastic Processes

3.1 Brownian Motion and Geometric Brownian Motion

Standard Brownian motion W_t, also known as the Wiener process, provides the mathematical foundation for modeling diffusive uncertainty in financial markets:

$$dW_t = \sqrt{dt} \cdot Z, \quad Z \sim \mathcal{N}(0,1)$$

where Z is a standard normal random variable.

Geometric Brownian Motion (GBM) adapts this framework for positive price processes $S_t > 0$:

$$dS_t = \mu S_t \, dt + \sigma S_t \, dW_t$$

where μ represents the drift coefficient and σ the volatility parameter.

Application of Itô's lemma yields the analytical solution:

$$S_t = S_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right) t + \sigma W_t\right)$$

The derivation proceeds by defining $f = \ln S_t$. Through Itô's lemma:

$$df = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW_t$$

Integration and exponentiation yield the final solution. Empirical research has validated GBM as particularly effective for modeling world crude oil prices, with minimal error between simulated and market prices [2].

3.2 Ornstein-Uhlenbeck Process for Mean Reversion

For commodities exhibiting mean-reverting characteristics (e.g., copper prices post-supply shock), the Ornstein-Uhlenbeck (OU) process provides a more appropriate modeling framework:

$$dX_t = \kappa(\theta - X_t) \, dt + \sigma \, dW_t$$

where κ represents the rate of mean reversion, θ the long-term equilibrium level, and σ the volatility.

The process has the following moment properties:

- Mean: $E[X_t] = \theta + (X_0 - \theta)e^{-\kappa t}$
- Variance: $\frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$

For practical implementation, discretization via the Euler-Maruyama method yields:

$$X_{t+1} = X_t + \kappa(\theta - X_t)\Delta t + \sigma\sqrt{\Delta t}Z$$

3.3 Jump-Diffusion Models

To capture discontinuous price movements arising from market shocks or unexpected events, Merton's jump-diffusion extension enhances the standard GBM framework:

$$dS_t = \mu S_t\,dt + \sigma S_t\,dW_t + S_t\,dJ_t, \quad J_t = \sum_{i=1}^{N_t}(Y_i - 1), \quad N_t \sim \text{Poisson}(\lambda t)$$

where N_t represents a Poisson process with intensity λ, and Y_i denotes the random jump size.

The corresponding Itô formula with jumps incorporates an additional term: $f(S^-(Y - 1))dN_t$.

4 The Mitsui & Co. Commodity Prediction Challenge

4.1 Competition Framework and Dataset

The Mitsui & Co. Commodity Prediction Challenge presented participants with a large-scale forecasting problem involving multiple time series from diverse financial markets [1]. The dataset comprises:

Table 1: Dataset Composition by Market

Market	**Count**	**Description**
LME (London Metal Exchange)	4	Metal commodities
JPX (Japan Exchange Group)	40	Japanese futures and equities
US (U.S. Securities Exchanges)	475	U.S. stocks and ETFs
FX (Foreign Exchange)	38	Currency pairs

The training data (train.csv) dimensions are 1917×558, with train$_{l}abels.csvat1917425.The424prediction$ $4days)withextendedevaluationperiodspost-October6, 2025.Participantsfacedcomputationalconstraintsi$

4.2 Evaluation Metric

The competition employed a variant of the Sharpe ratio as the evaluation criterion:

$$\text{Score} = \frac{E[\rho(\hat{y}, y)]}{\sigma[\rho]}$$

where $\rho(\hat{y}, y)$ represents the Spearman correlation between predictions and actual values. This metric emphasizes stability of predictive performance over time by rewarding consistent rank correlations while penalizing high variance in prediction quality.

The top solution achieved a Sharpe ratio of 1.341129, representing a 941.8% improvement over the baseline approach through comprehensive coverage of all targets and lags combined with advanced feature engineering and GPU acceleration [1].

5 Machine Learning Pipeline: Ensemble Tree Boosters

5.1 Pipeline Architecture

The implemented machine learning pipeline adopts a multi-model ensemble approach to address the competition's forecasting challenges. The architecture comprises several key components:

- **Data Preprocessing**: Comprehensive feature engineering including lagged and rolling features [1], date-derived temporal features (dayofweek, month, etc.), and robust missing value handling.
- **Model Configuration**: Multiple gradient boosting implementations (LightGBM, XGBoost, CatBoost) with conservative hyperparameters (learning rate=0.005, early stopping=20, maximum rounds=200) to prevent overfitting.
- **Validation Strategy**: K-fold cross-validation (n=5) with temporal awareness to address time series dependencies.
- **Ensemble Mechanism**: Weighted averaging of predictions across folds and model types to enhance stability and generalization.

The pipeline processes each of the 424 targets independently, concatenating training features with corresponding labels and identifying each series through target$_{i}didentifiers$.

5.2 Feature Engineering Strategy

Effective feature engineering proved critical to success in similar forecasting challenges [5]. The implemented strategy includes:

This comprehensive approach to feature engineering aligns with strategies employed by top performers in similar competitions, where creative feature transformation often differentiates leading solutions [5].

Table 2: Feature Engineering Components

Component	**Description**
Temporal Features	Cyclical encoding of date components (day of week, month), time-to-holiday indicators, and quarterly seasonality markers.
Statistical Features	Rolling window statistics (mean, standard deviation, min/max) across multiple lookback periods to capture local trends.
Cross-Asset Features	Pairwise correlations and ratios between economically related assets to capture inter-market dependencies.
Lag Features	Price and return lags at economically meaningful intervals (1, 5, 20 periods) to capture autoregressive patterns.

5.3 Training Performance and Analysis

The ensemble model demonstrated consistent convergence across validation folds:

Table 3: Training Performance Across Models

Model	**Fold 0 RMSE**	**Fold 1 RMSE**	**Final RMSE**
LightGBM	0.030015 → 0.030014	0.029864 → 0.029848	0.02993
XGBoost	Stagnation 0.03001-0.02985	Similar pattern	0.02993
CatBoost	Convergence comparable to LGBM	Slightly higher variance	0.02995

Notably, LightGBM achieved early stopping at iteration 43 for Fold 0, while Fold 1 utilized the full 200 training rounds, suggesting variability in convergence patterns across different temporal segments. The XGBoost model exhibited stagnation in later training stages, potentially due to the conservative learning rate limiting parameter updates. The final ensemble achieved an average RMSE of approximately 0.0298 across all targets, with low cross-validation variance indicating stable performance suitable for the competition's evaluation metric.

The pipeline efficiently handled the high-dimensional input space (565 features after engineering) through strategic use of shallow tree architectures (LightGBM: num$_{l}eaves = 8; XGBoost : max_{d}epth = 4), balancingmodelcomplexitywithcomputationalconstraints$.

6 Empirical Validation and Hybrid Framework

6.1 Exploratory Data Analysis

Exploratory analysis revealed several important characteristics of the commodity data:

Augmented Dickey-Fuller tests confirmed mean reversion in certain series (p¡0.05 for copper spreads), while correlation analysis revealed strong relationships between economically linked assets (XOM-XLE=0.85; LQD-EEM=-0.62). The observed excess kurtosis across multiple series (4.1-5.8 versus 3.0 for GBM) suggests that jump components may be necessary to fully capture commodity price dynamics.

Table 4: Distributional Properties Across Markets

Series	**Mean**	**Volatility**	**Skewness**	**Kurtosis**
LME Copper	0.0008	0.018	-0.14	4.1
Target_0	-0.0012	0.023	0.31	5.8
GBM Simulation	0.0002	0.016	0.02	3.0

6.2 Hybrid Stochastic-ML Framework

To leverage the complementary strengths of stochastic modeling and machine learning, we propose a hybrid framework that augments traditional feature sets with simulated paths from calibrated stochastic processes:

- **OU Process Features**: Simulated mean-reverting paths applied to LME and JPX series to capture reversion dynamics
- **Jump-Diffusion Signals**: Conditional volatility estimates from Merton-style models to identify regime change periods
- **GBM Uncertainty Bands**: Multiple simulated trajectories to quantify forecast uncertainty

These stochastic features serve as additional inputs to the ensemble tree models, creating a rich representation that combines empirical patterns with theoretical priors. The approach draws inspiration from singular spectrum analysis techniques that have successfully improved ARIMA forecasts through decomposition into stochastic flows [4].

Risk simulation through Monte Carlo methods $(10^4 paths, jump-diffusion) produces Value-at-Risk estimates$ ($\text{VaR}_{95\%} \approx -2.5\%$ for copper) that complement point forecasts from the machine learning pipeline, providing a more comprehensive risk assessment framework.

7 Conclusion and Future Research

This research demonstrates the complementary value of stochastic process modeling and machine learning for commodity price forecasting. The mathematical rigor of SDEs provides theoretical grounding and interpretability (e.g., κ parameters quantifying mean reversion rates), while ensemble tree methods offer scalable pattern recognition from high-dimensional datasets (achieving RMSE¡0.03 across 424 targets).

The proposed hybrid framework, integrating simulated stochastic features with traditional machine learning inputs, represents a promising direction for achieving both accuracy and stability in financial forecasts. This approach aligns with recent advances in decomposition methods that enhance traditional forecasting models [4] and reflects the feature engineering sophistication that differentiates top performers in forecasting competitions [1].

Future research directions include:

- Implementation of Hawkes processes to capture self-exciting volatility clustering

- Online learning techniques for real-time model adaptation to changing market regimes
- Application of singular spectrum analysis for more sophisticated decomposition of price series into constituent stochastic flows
- Integration of exogenous macroeconomic variables and news sentiment data

The rapid evolution of commodity forecasting methodologies, driven by both academic research and competitive platforms like Kaggle, continues to enhance our ability to navigate complex financial markets with improved precision and stability.

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